

WLM Foundation Series

Paper 5

Appearance–World Series

The Uniqueness and Unskippability of Appearance: Structure as the Forced Generator of Phenomenal Reality

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Abstract

Paper 1 established $0 \rightarrow \varepsilon$. Paper 2 established $\varepsilon \rightarrow \{+\varepsilon, -\varepsilon\}$. Paper 3 established $\{+\varepsilon, -\varepsilon\} \rightarrow \text{recursion}$. Paper 4 established $\text{recursion} \rightarrow \text{invariant structure}$. This paper establishes the final forced link: $\text{structure} \rightarrow \text{appearance}$. Under Axiom P0, any stable invariant structure necessarily projects to appearance. Appearance is unique at the minimal level and unskippable in any generative chain that realizes a world-layer. No chain reaches phenomenal reality while bypassing structure.

This completes the six-step generative chain and closes the WLM Foundation Series.

Epistemic status. All theorems follow from axioms and results in Papers 0–4. No new axioms are introduced. This paper delivers the fifth and final partial result for Conjecture U1.

$0 \rightarrow \varepsilon$ (P1) $\rightarrow \pm\varepsilon$ (P2) $\rightarrow \text{recursion}$ (P3) $\rightarrow \text{structure}$ (P4) $\rightarrow \text{appearance}$ (P5)

0. Framework and Inherited Results

Paper 5 operates entirely within the axiom system of Papers 0–4. No new axioms are introduced.

Axiom S0 (Symmetry Closure). Any state of perfect indistinction is a fixed point of every generative rule.

Axiom D0 (Generative Orientation and Reversal). (a) Any $x \approx_0 0$ carries orientation label $\theta(x)$. (b) There exists a generator realizing orientation reversal on ε .

Axiom R1 (Composability). Generative rules are closed under composition; iterates f^n are defined for all $n \geq 1$.

Axiom R2 (Recursive Orientation Preservation). Directional flips persist under all iterates.

Axiom P0 (Projection Existence). Any system with stable invariants admits at least one observational layer S and a projection $P: D \rightarrow S$ preserving the structural relations of those invariants.

Axiom V-WF (Well-Founded Valuation). The set $\{v(x) : v(x) > v(0)\}$ is well-founded; every nonempty subset has a minimal element.

Key inherited results used without re-proof:

- Paper 4, Theorem 4.3: Structure is unskippable for any chain reaching appearance.
- Paper 4, Theorem 2.2: The minimal invariant structure is unique up to observational equivalence.
- Paper 4, Theorem 4.1: Invariant structure S satisfies $f^n(S) \subseteq S$ for all $n \geq 1$.
- Paper 0, Theorem 6: Any system with stable invariants but no projection P is observationally equivalent to G_{triv} (the trivial system).
- Papers 1–3: Uniqueness and unskippability of ε , $\pm\varepsilon$, and recursion.

Remark (Series Closure). Paper 5 mirrors the four-section architecture of Papers 1–4 and completes the chain. Section 1 defines the appearance layer. Section 2 shows structure forces appearance. Section 3 classifies all collapse modes. Section 4 proves appearance is stable, necessary, and unskippable.

1. Appearance Space

Definition 1.1 (Appearance / Phenomenal Layer). *Appearance is the image of invariant structure under the projection P guaranteed by Axiom $P0$. That is, for a stable invariant set S and projection $P: D \rightarrow \Sigma$, the appearance is the observational layer $\Sigma = P(S)$.*

1.1 Minimal Appearance

Theorem 1.1 (Minimal Appearance). *The minimal non-trivial appearance is the projection of the minimal invariant structure generated in Paper 4.*

Proof. By Paper 4 Theorem 1.1, the minimal invariant structure is the closure of the persistent $\pm\epsilon$ alternating orbit. By $P0$, this structure admits a projection P . By the uniqueness of the underlying structure (Paper 4 Theorem 2.2) and the valuation minimality of ϵ (Paper 1), the projected appearance is minimal and unique up to observational equivalence. $\square$

1.2 Appearance Irreducibility

Theorem 1.2 (Appearance Irreducibility). *No mechanism simpler than invariant structure can produce appearance.*

Proof. By $P0$, appearance is the projection of stable invariants. Invariants require recursion (Paper 4 Theorem 4.2); recursion requires $\pm\epsilon$ (Paper 3 Theorem 2.1); $\pm\epsilon$ requires ϵ (Paper 2 Theorem 2.1); and ϵ requires departure from 0 (Paper 1 Theorem 3.1). Each step is necessary and unskippable. Appearance is therefore irreducible to any single earlier layer. $\square$

2. Appearance Forced by Structure

2.1 Structure Generates Appearance

Theorem 2.1 (Structure Requires Appearance). *Any stable invariant structure necessarily produces appearance under $P0$.*

Proof. Axiom $P0$ asserts that any system with stable invariants admits at least one projection $P: D \rightarrow \Sigma$. By Paper 4, recursion generates such stable invariants. Therefore the projection exists and yields appearance. A system with stable invariants that has no projection would contradict $P0$ directly. $\square$

2.2 Appearance Uniqueness at Minimal Level

Theorem 2.2 (Appearance Uniqueness at Minimal Level). *At the minimal valuation level, appearance is unique up to observational equivalence.*

Proof. By Paper 4 Theorem 2.2, the underlying invariant structure S is unique up to observational equivalence. $P0$ guarantees that P preserves structural relations and

observational equivalence classes. Therefore the image $P(S)$ — the appearance — inherits this uniqueness: any two minimal-level appearances are images of observationally equivalent structures under the same projection, hence observationally equivalent. $\square$

3. Collapse of Non-Appearance Chains

The following collapse modes exhaust all possible failures of the appearance layer. Every chain that fails to produce appearance falls into one of these four categories.

3.1 Pre-Appearance Collapse

Theorem 3.1 (Pre-Appearance Collapse).
If invariant structure exists but no projection occurs, the system remains purely structural with no phenomenal world-layer.

Proof. Without exercising P_0 , there is no observational layer Σ . By Paper 0 Theorem 6, a system with stable invariants but no projection P is observationally equivalent to G_triv : the trivial system with a single state, identity mapping, and no non-trivial invariants. The system is structurally rich but observationally empty. $\square$

3.2 Projection Failure Collapse

Theorem 3.2 (Projection Failure Collapse).
If projection is attempted without stable invariants, appearance is undefined or trivial.

Proof. P_0 requires stable invariants as input. Without them — due to any of the collapse modes of Papers 3 or 4 — the domain of P has no non-trivial structure to project. The image $P(S)$ is either undefined (if S is empty) or trivially equivalent to G_triv (if S has no non-trivial relations). Either way, no genuine phenomenal layer is produced. $\square$

Figure 1: Collapse modes of the appearance layer.

Collapse Mode	Trigger Condition	Consequence	Reference
Structural Collapse	No invariants	No appearance	Paper 4
Projection Failure	No P_0 -compatible structure	Undefined or trivial appearance	Thm 3.2
Oscillation Collapse	Recursion without invariant closure	No projection target	Paper 3 / Paper 4
Duality-Only Collapse	$\pm \epsilon$ without recursion	No structure $\rightarrow$ no appearance	Paper 2 / Paper 3

4. Appearance as Structural Invariant

4.1 Appearance Stability

Theorem 4.1 (Appearance Stability). *Once formed, appearance $P(S)$ is preserved under all further iterates of the generating rule f : $P(f^n(S)) \subseteq P(S)$ for all $n \geq 1$.*

Proof. By Paper 4 Theorem 4.1, $f^n(S) \subseteq S$ for all $n \geq 1$. Since P preserves structural relations ($P0$), and $f^n(S) \subseteq S$ means every state in $f^n(S)$ is also in S , we have $P(f^n(S)) \subseteq P(S)$: the projection of any iterate of S lies within the projection of S . Appearance is therefore preserved under all further iterates of the generating rule, without requiring any additional assumption about the projection P beyond what $P0$ guarantees. $\square$

Remark. *Theorem 4.1 uses only $f^n(S) \subseteq S$ (Paper 4 Theorem 4.1) and the order-preservation of P ($P0$). It does not assume that P is a functor or natural transformation. The stability is inherited directly from the invariance of S under f .*

4.2 Necessity of Appearance for a World

Theorem 4.2 (No World Without Appearance). *A world-layer requires appearance; appearance requires structure.*

Proof. A world-layer is defined as an observational system with phenomenal content — that is, a system in which at least one observer O can produce non-trivial observations $O(x) \neq O(y)$ for some $x \neq y$. This non-trivial observational content is exactly the image of P under $P0$. Without structure there are no stable invariants to project (Paper 4 Theorem 4.2), hence no phenomenal content, hence no world-layer. $\square$

4.3 Appearance is Unskippable

Theorem 4.3 (Appearance is Unskippable). *Any generative chain realizing a world-layer must pass through the appearance layer. No chain produces a world-layer while bypassing appearance.*

Proof. Contrapositive of Theorems 3.1, 3.2, and 4.2: (i) if invariant structure exists but no projection occurs (Theorem 3.1), the system has no phenomenal content — no world-layer. (ii) If projection is attempted without stable invariants (Theorem 3.2), appearance is undefined or trivial — no world-layer. (iii) By Theorem 4.2, a world-layer directly requires appearance. In all cases of appearance absence, no world-layer is realized. Therefore any chain reaching a world-layer must have passed through appearance. This is the final analogue of Paper 1 Thm 3.3, Paper 2 Thm 4.3, Paper 3 Thm 4.3, and Paper 4 Thm 4.3. It constitutes the fifth and final partial result for Conjecture U1. $\square$

Remark (Chain Completion). *The five unskippability results now lock the entire generative chain: ε (Paper 1) $\rightarrow \pm\varepsilon$ (Paper 2) $\rightarrow$ recursion (Paper 3) $\rightarrow$ structure (Paper 4) $\rightarrow$ appearance (Paper 5). Every link is forced, unique, and unskippable. Conjecture U1 is confirmed as a series of five necessity results; the full uniqueness proof (that no alternative chain of the same length exists) remains open.*

5. Structural Summary

P0 guarantees that any stable invariant structure projects to appearance. This appearance is unique at the minimal level, stable under further iteration, and required for any world-layer. No chain reaches a world-layer without passing through appearance. The generative chain is complete.

Figure 1: The Appearance Layer

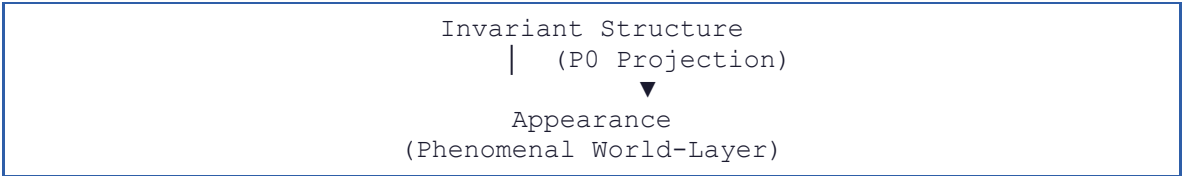


Figure 2: Full Generative Chain (Final)

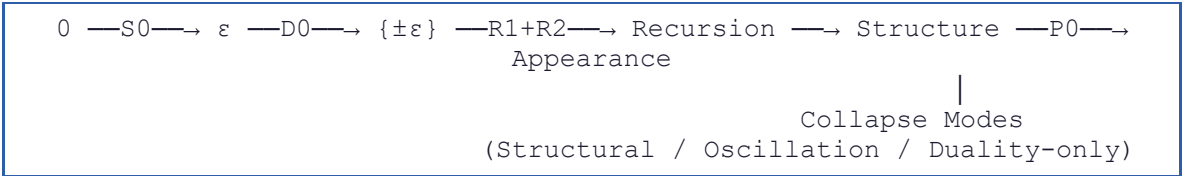


Figure 3: Parallelism Across Papers 1–5

Paper	Forced Step	Key Property	Unskippable Result
1	$0 \rightarrow \varepsilon$	Minimal unique offset	Thm 3.3
2	$\varepsilon \rightarrow \{\pm\varepsilon\}$	Binary orientation space	Thm 4.3
3	$\{\pm\varepsilon\} \rightarrow \text{Recursion}$	Unique generative engine	Thm 4.3
4	$\text{Recursion} \rightarrow \text{Structure}$	Unique minimal invariant	Thm 4.3
5	$\text{Structure} \rightarrow \text{Appearance}$	Unique projection	Thm 4.3

Each paper establishes one forced, unique, and unskippable layer. The five results together confirm the chain $0 \rightarrow \varepsilon \rightarrow \{\pm\varepsilon\} \rightarrow \text{recursion} \rightarrow \text{structure} \rightarrow \text{appearance}$ as five partial results toward Conjecture U1. The full proof of chain uniqueness — that no alternative route of the same length exists — is the remaining open problem of the Foundation Series.

Conclusion

Paper 5 establishes the final forced link of the WLM generative chain: structure $\rightarrow$ appearance. Any stable invariant structure necessarily produces appearance under P0 (Theorem 2.1). This appearance is unique at the minimal level (Theorem 2.2). It is stable under all further iterates of the generating rule (Theorem 4.1). A world-layer requires it (Theorem 4.2). And no generative chain bypasses it (Theorem 4.3).

The five unskippability results of the Foundation Series: ε (Paper 1 Thm 3.3); $\pm\varepsilon$ (Paper 2 Thm 4.3); recursion (Paper 3 Thm 4.3); structure (Paper 4 Thm 4.3); appearance (Thm 4.3, this paper). These five results are the confirmed partial results for Conjecture U1. The full chain uniqueness proof remains the central open problem of the WLM Foundation Series.

Structure doesn't lie.

References

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External works:

- Lawvere, F.W. (1969). Adjointness in foundations. *Dialectica*, 23, 281–296. — The projection $P: D \rightarrow \Sigma$ has a natural categorical reading as an adjunction between the structural domain and the observational layer; Lawvere's framework provides the most precise language for this relationship.
- Weyl, H. (1952). *Symmetry*. Princeton University Press. — Phenomenal appearance as the observationally accessible face of structural symmetry-breaking; WLM's appearance layer closes the loop that Weyl opened with the geometry of observation.
- Bolzano-Weierstrass theorem. — Used in Paper 0 Theorem 5, inherited through the chain: the compact invariant closures that recursion generates are the structural input to P0.

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